

# RAG-NIMBF: A Non-Invasive, Modular Benchmarking Framework for Automated RAG Configuration Analysis

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## Problem

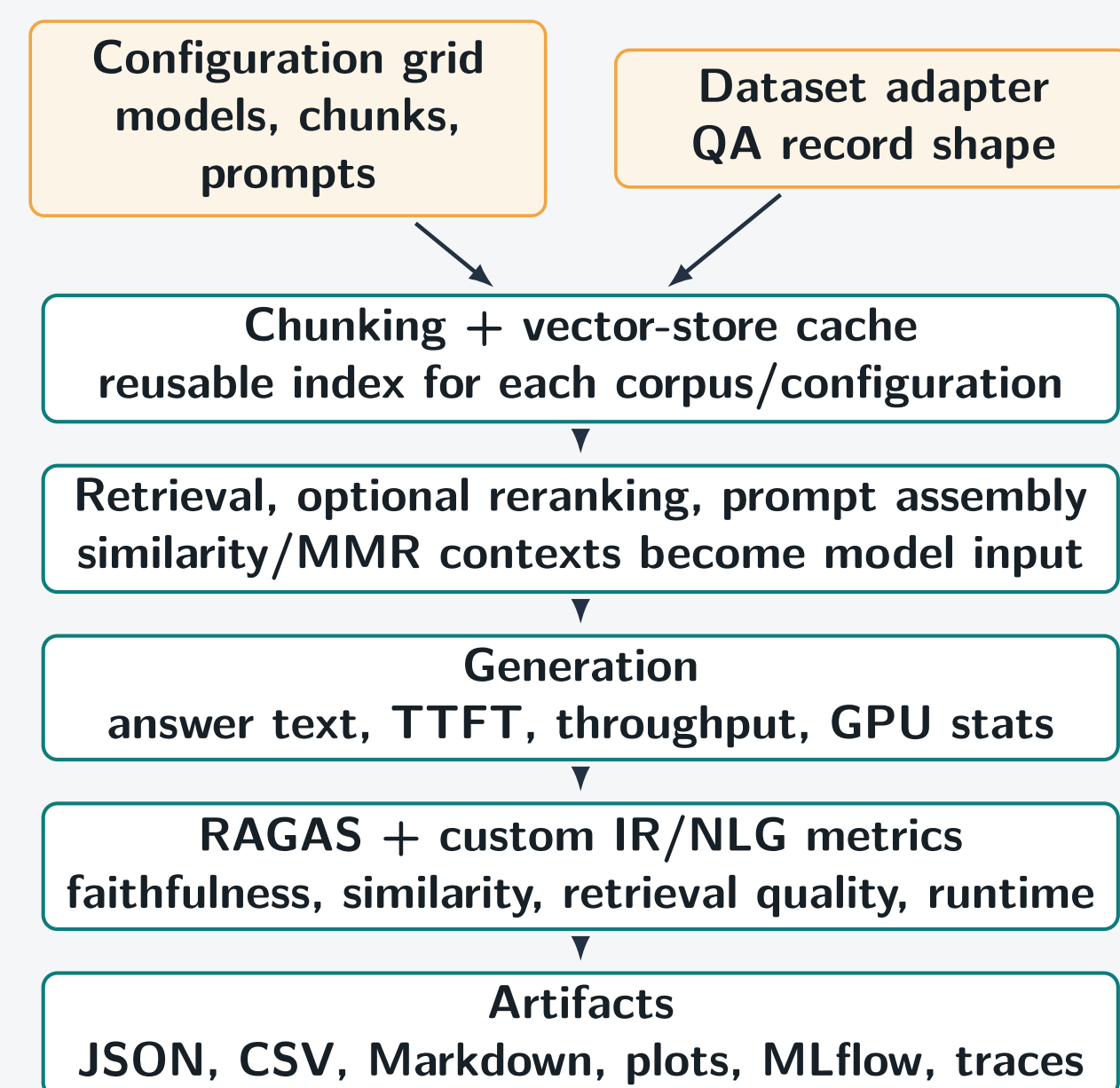
Retrieval-augmented generation (RAG) systems expose a large configuration space. Chunking, overlap, embedding model, vector database, retrieval strategy, reranking, prompt template, generator model, and evaluator choice can all move quality and run-time in different directions.

**Research goal:** make RAG configuration choices measurable, repeatable, and comparable without rewriting the target pipeline for each experiment.

## Framework Contribution

- Environment-driven benchmark grids for local and OpenAI-compatible model providers.
- Dataset adapters normalize QA corpora into one benchmark record shape.
- Reusable retrieval core supports Chroma and LanceDB vector stores, similarity search, MMR, optional HyDE, and cross-encoder reranking.
- Generation layer captures answer text, time to first token, throughput, and optional GPU statistics.
- Reporting exports JSON, CSV, Markdown, static plots, MLflow runs, and optional tracing artifacts.

## Pipeline



## Experiment Space

| Dimension  | Current values in project                    |
|------------|----------------------------------------------|
| Chunking   | recursive, semantic, text, token, markdown   |
| Chunk size | 500, 1000, 1500 and variants                 |
| Overlap    | 100, 200, 350 and variants                   |
| Retrieval  | similarity, MMR                              |
| Reranking  | none, MiniLM cross-encoder                   |
| Embedding  | nomic-embed-text:latest snapshot             |
| Datasets   | SQuAD / WikiQA-style QA adapters             |
| Models     | Qwen3, Qwen3.5, Qwen3.6, GPT-OSS, GPT-4o-mir |

## Reproducibility Controls

- Config grid:** environment-defined combinations generate every run.
- Run folders:** each benchmark writes to `results/runN/`.
- Vector cache:** corpus fingerprints avoid accidental re-embedding.
- Stage timings:** indexing, retrieval, generation, and evaluation are timed.
- Artifact trail:** JSON, CSV, plots, MLflow, and traces preserve evidence.

## Observed Answers

- Prompting:** across  $N \geq 100$  rows, detailed prompts raise mean faithfulness (0.908,  $n = 125$ ) over concise (0.838,  $n = 132$ ), while BERT-F1 stays nearly unchanged (0.873 vs. 0.871).
- Retrieval:** similarity is stronger for faithfulness (0.886,  $n = 146$ ) vs. MMR (0.854,  $n = 111$ ), while MMR improves nDCG@5 (0.779 vs. 0.686) and recall@5 (0.826 vs. 0.559).
- Chunking:** recursive and semantic chunking are close on mean faithfulness (0.875,  $n = 158$  vs. 0.867,  $n = 99$ ), while semantic improves mean nDCG@5 (0.747 vs. 0.713).
- Winner:** no single configuration dominates: faithfulness, BERT-F1, nDCG@5, recall@5, METEOR, and runtime select different runs.

## Evaluation Lenses

- LLM-judged quality:** faithfulness, context recall, semantic similarity, and related RAGAS fields where available.
- Retrieval quality:** hit@k, nDCG@k, recall@k, context relevance, with heuristic or gold-document modes.
- Answer similarity:** ROUGE-L, BLEU, METEOR, and optional BERTScore precision, recall, F1.
- Performance:** total runtime, stage timings, time to first token, tokens per second, and GPU usage when available.

## Prompt Templates

### Concise Answer Prompt

Answer the question using ONLY the provided context.  
RULES:  
- Give a short, direct, factual answer - one sentence maximum.  
- Do NOT explain your reasoning, add context, or use markdown formatting.  
- Do NOT say 'Based on the context' or similar preamble.  
- If the context does not contain the answer, say 'I cannot answer from the provided context.'  
Examples:  
Q: What circuit court is Maryland? → The Circuit Courts of Maryland are the state trial courts of general jurisdiction.  
Q: How many members are in the House of Representatives? → The total number of voting representatives is fixed at 435.  
Q: Who wrote the music for Star Wars? → John Williams.

### Detailed Answer Prompt

Answer the question based only on the provided context. Provide a clear, complete answer in full sentences. Include relevant calculations or reasoning steps when applicable. Do not start your answer with phrases like 'Based on the provided context' or 'Based on the given context'. Answer directly.

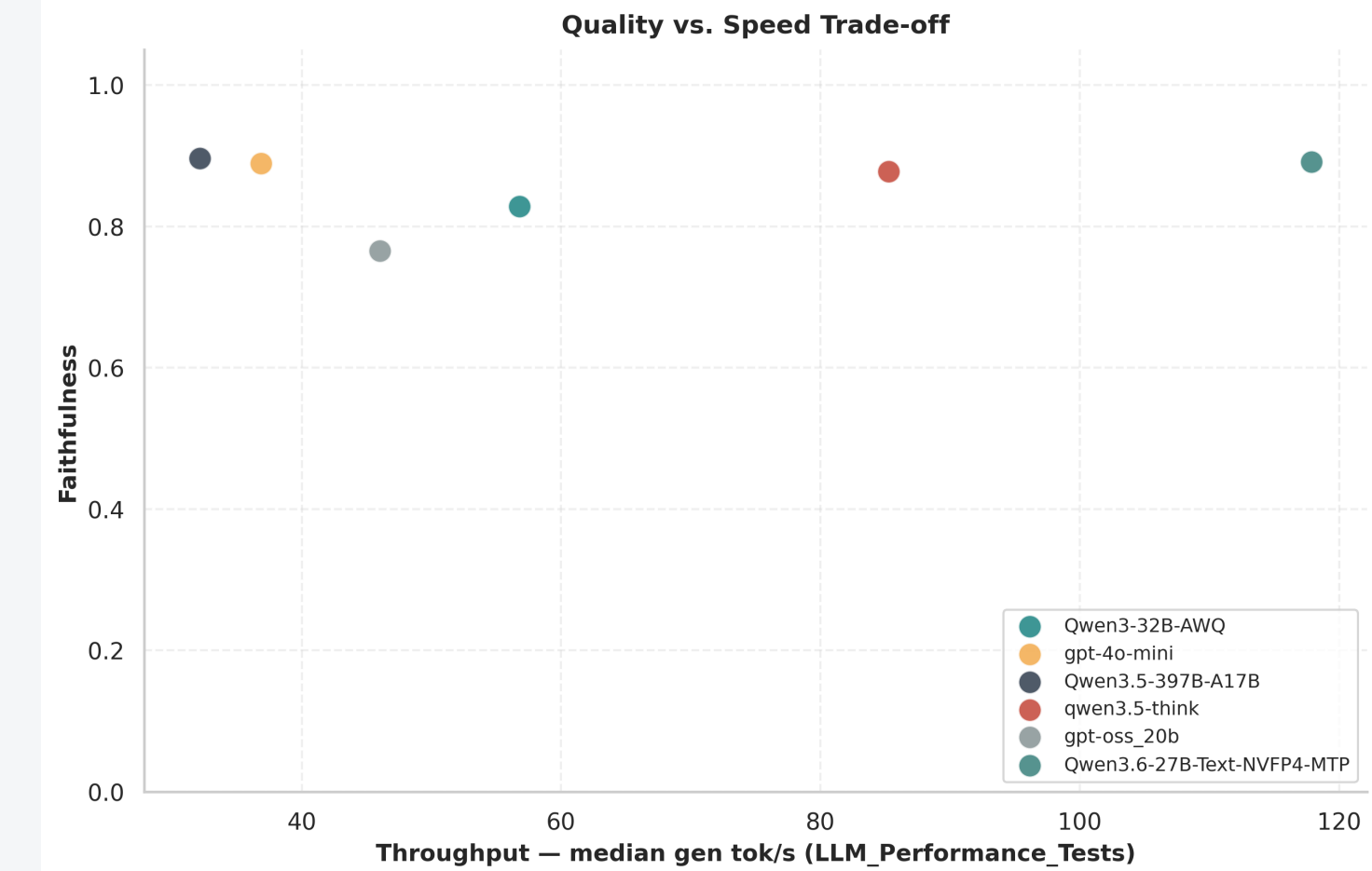
## Current Results Snapshot

The current result corpus contains 274 JSON configuration rows across 86 benchmark reports, with 257 rows at  $N \geq 100$ . The manual evaluation matrix tracks configuration IDs up to 222 and currently marks 212 full  $N = 100$  rows as tested. The table below highlights the strongest observed configurations for three evaluation lenses, demonstrating that different configurations can excel in different metrics, underscoring the importance of multi-metric evaluation in RAG benchmarking.

| Observed strongest configuration                          | Faith. | BERT-F1 | nDCG@5 | N   |
|-----------------------------------------------------------|--------|---------|--------|-----|
| Qwen3.5-397B, detailed, recursive 500/100, similar-ity    | 0.959  | 0.835   | 0.702  | 300 |
| Qwen3-32B-AWQ, concise, semantic, MMR                     | 0.871  | 0.969   | 1.000  | 100 |
| deepseek-v4-pro, concise, recursive 1000/100, simi-larity | 0.845  | 0.919   | 0.675  | 100 |

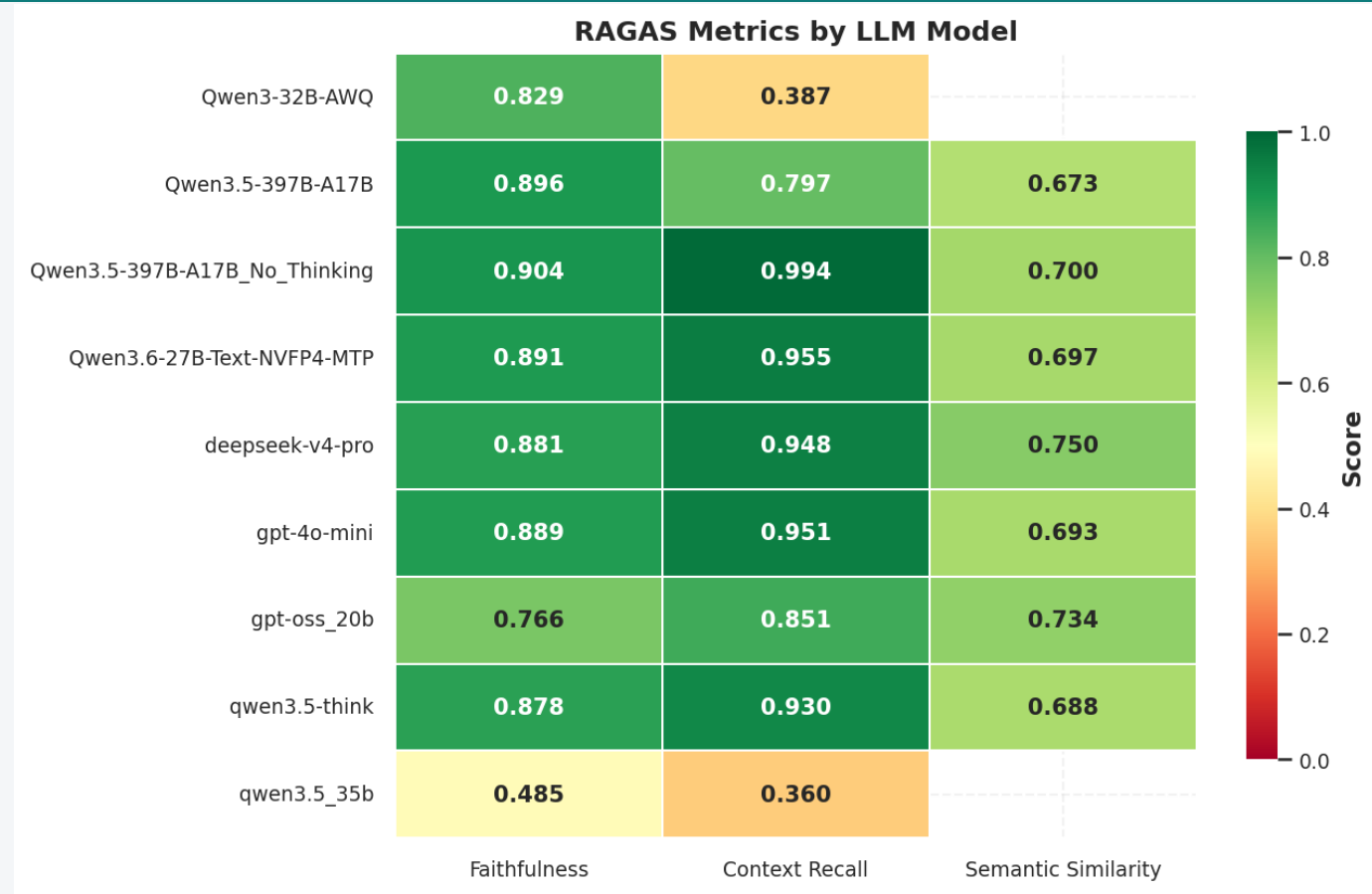
**Interpretation:** the best run depends on the metric. Strong faithfulness, strong retrieval ranking, and strong lexical/semantic answer similarity do not always select the same configuration.

## Quality vs. Speed

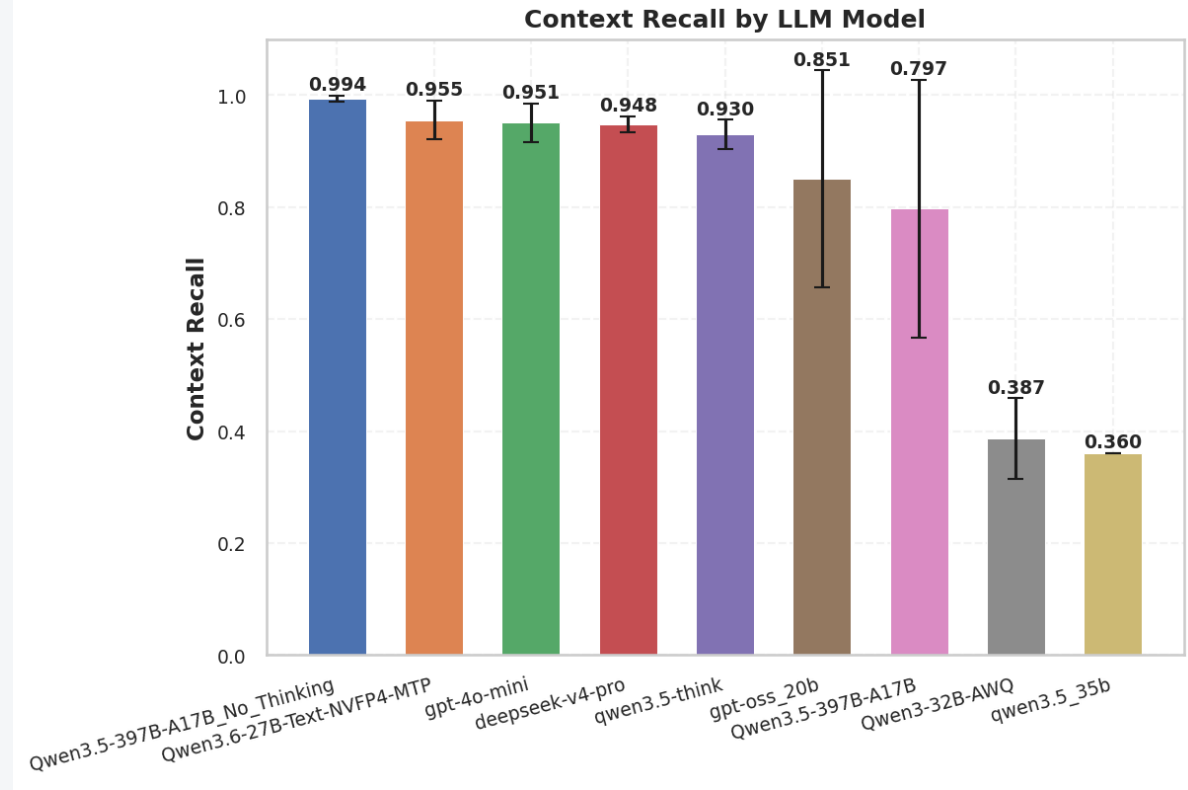


Faithfulness vs. throughput; dashed line = Pareto frontier. No single config dominates — frontier trades quality for speed across LLMs.

## Prompt and Model Effects



## Retrieval Trade-Offs



MMR can improve diversity-sensitive retrieval metrics in some runs, while plain similarity can preserve higher faithfulness in others. Reranking helps in selected settings, but adds latency and should be judged jointly with answer quality.

## LLM Runtime Baseline

| Model / backend            | Latency | Gen tok/s |
|----------------------------|---------|-----------|
| Qwen3.5-397B / Cloud       | 7.82s   | 130.9     |
| Qwen3.5-Think / Cloud      | 7.84s   | 130.6     |
| Qwen3.6-27B / RTX5090 vLLM | 8.67s   | 118.1     |
| GPT-4o-mini / Cloud        | 10.12s  | 101.1     |
| Qwen3-32B / RTX5090 vLLM   | 17.36s  | 59.0      |
| gpt-oss-20B / Spark Ollama | 18.13s  | 56.5      |

Independent LLM performance baseline from LLM\_Performance\_Tests-main: sequential single-call runs with  $N = 1$ , max output 1024 tokens. Throughput is generated output tokens per second. TTFT is omitted from the table because two traces are non-streamed; output-token count is a fixed benchmark cap rather than a differentiating metric.

## Limitations

- Current results are local, hardware-dependent, and partly duplicated by reruns.
- SQuAD/WikiQA-style corpora may not transfer to domain RAG workloads.
- LLM evaluator reliability, stochastic generation, and provider changes remain validity threats.
- Manual matrix entries must be reconciled with generated result files before publication.

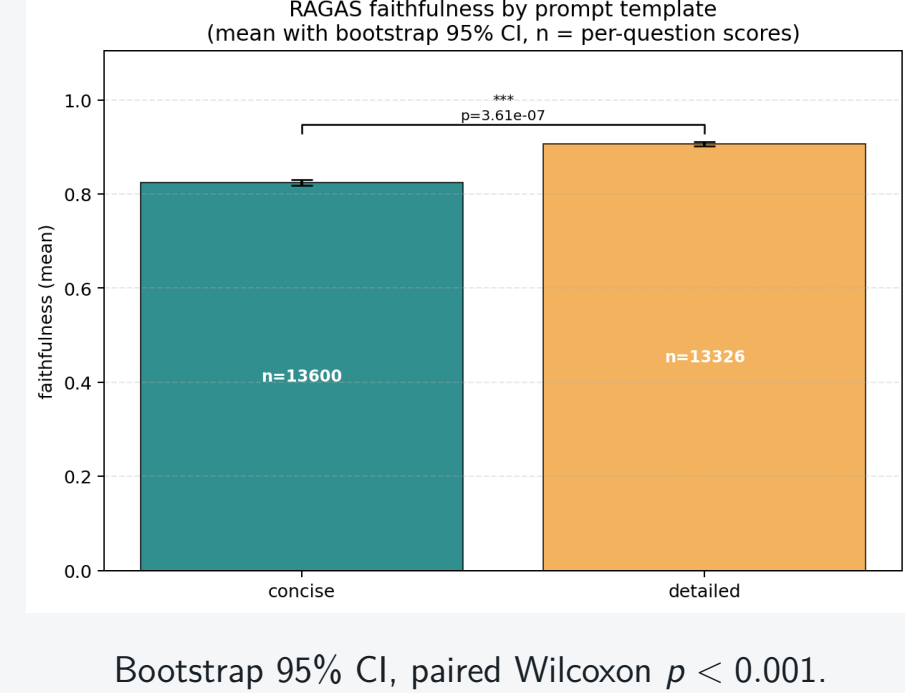
## Takeaways for Practitioners

- Benchmark complete RAG configurations, not isolated components.
- Report multiple metrics because each metric rewards different behavior.
- Store machine-readable artifacts for every run.
- Treat prompt template and retrieval strategy as first-class tuning variables.
- Keep limitations visible when using LLM-judged metrics.

## Future Work

- Datasets:** extend beyond SQuAD/WikiQA to domain corpora (legal, medical, enterprise KB).
- Scale:** raise  $N$  per config and add significance tests so single-run noise is visible.
- Retrieval ablation:** full HyDE and cross-encoder sweep, plus hybrid lexical+dense retrieval.
- Backends:** compare Chroma vs. LanceDB on index time, memory, and query latency at fixed corpus size.

## Statistical Robustness



## References

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